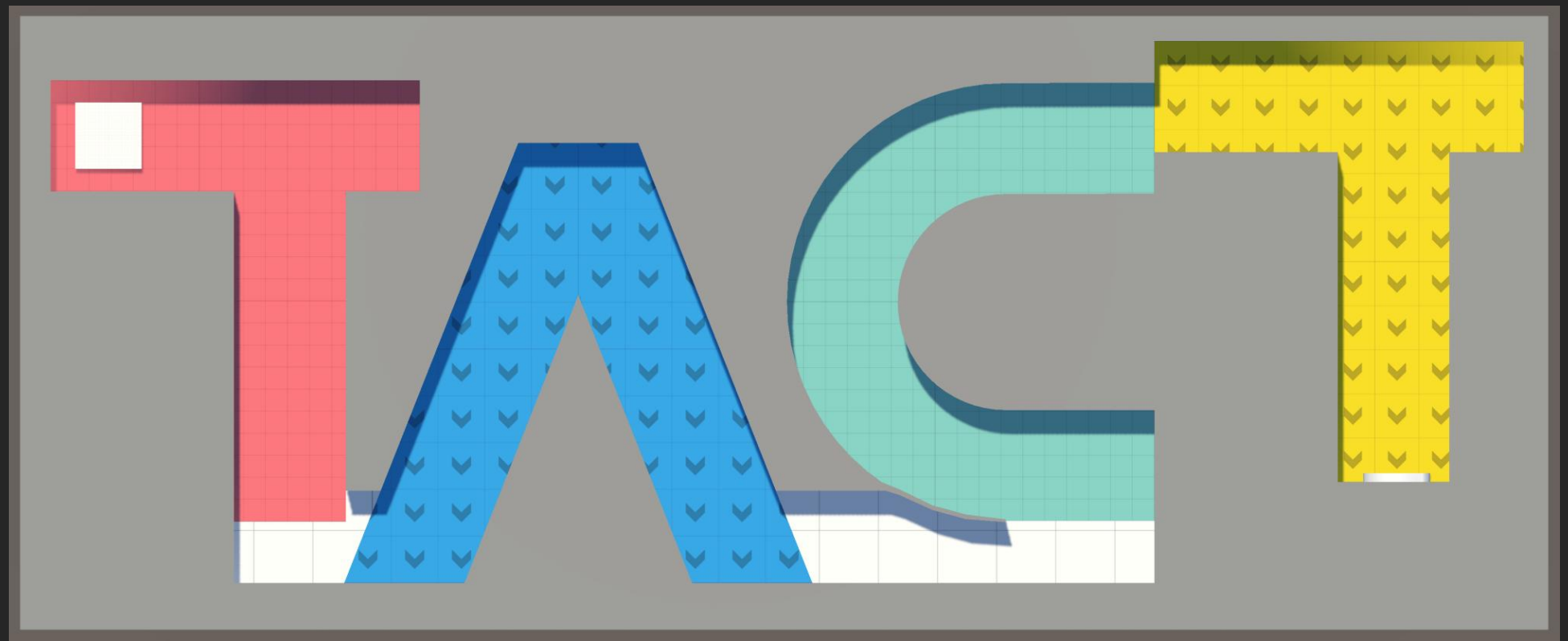
Introduction to Robotics

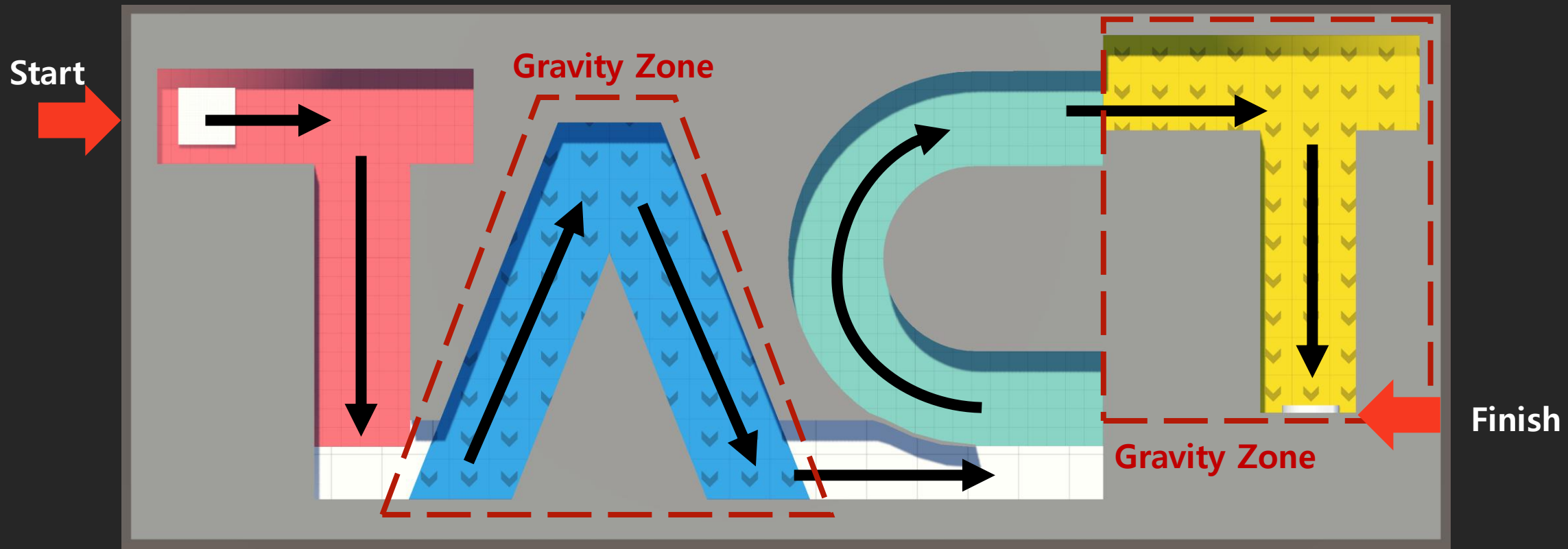
TA session 03

Week 7. Project1 Q&A



Project 1: Teleoperation

- Building a spaceship controlled by human, without sensors.



Files to be submitted

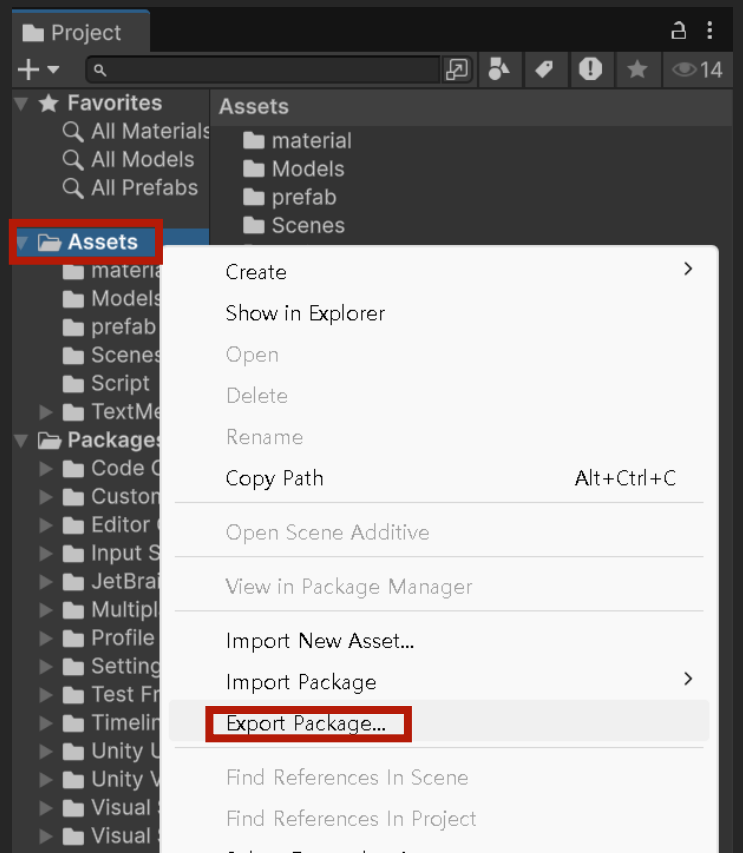
Due: 10/20(Mon) 18:00

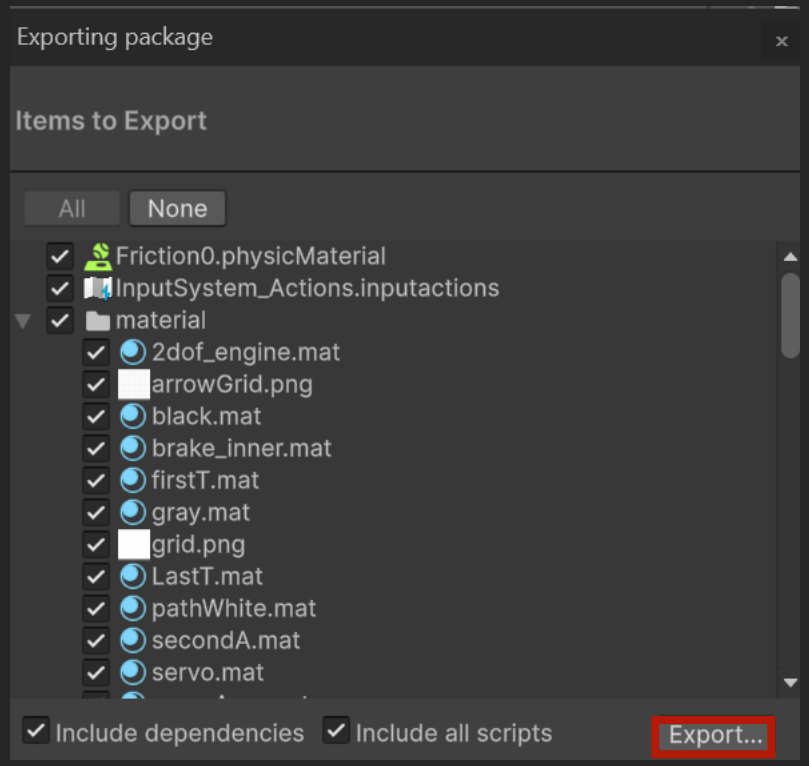
1. **Performance report** (template announced in blackboard)
   - Performance record
   - Participation details
   - Discussion
2. **Performance video**
   - Proof of your performance report.
   - full view of Unity scene, game window, controller and hands, recorded from starting line to finish line.
3. **Resource file Unity package**
   - Must contain full assets of the unity project

- If manipulation is suspected in the submitted file, additional session may be required to demonstrate the control.

How to make unitypackage file

- Project window
  - right click Assets folder > Export Package > Export





Evaluation criteria

- **Performance Report** (10 points, Team score)
  - Final Time = Time + Number of Collisions * 5 seconds
  - If a collision occurs, no further collisions will be detected for 5 seconds.
    - 200 seconds or more: 2 points
    - 200-170: 4 points
    - 170-140: 6 points
    - 140-110: 8 points
    - 110 seconds or less: 10 points
  - **The highest-scoring team: 5 additional points will be awarded to the final grade.**
    - This team's spaceship design, control system code, and video will be made public.

Evaluation criteria

- **Presentation** (15 minutes, 15 points, Individual score)
  - Presentation schedule is announced on the blackboard.
    - 10/21 (Tue): teams 1-6
    - 10/23 (Thu): teams 7-11
  - Each team will present their performance report in the professor's office.
  - Each team member present their role **(All team members must present)**
    - Performance record
    - Controller design
    - Rocket design
    - Control system design
  - **If you are not prepared for your presentation, you must attend the additional session on 10/23 from 19:00 to 22:00 for an additional presentation.**

Restrictions

If any of the restrictions are violated, the performance report will be disqualified. (0 points).

- **Overall**
  - Do not drive off the given track (through a wall or across the y-axis)
  - If you want to use external libraries or assets, be sure to get prior approval.
  - You must not modify the structure, size, or components of the track or time, physics system in Unity.
- **Spacecraft**
  - Total size must fit in 9 * 9 * 5m box.
  - Only one body part can be used, and it must be the highest parent in the hierarchy.
  - Do not manipulate the size, weight or performance of the parts. (except frames)
  - The total thrust of all engines used in a rocket must be less than 60.
  - A completed spacecraft must not have any disconnected parts.
  - Except for the frame, no parts can overlap.
    - Frame-frame (o)
    - Frame-part (o)
    - Part-part (x)
  - Design so that interference does not occur even when the part is rotated by a servo motor.
- **Controller**
  - Only the electronic components provided should be used in the controller.
    - Allows use of non-electronic components not provided
- **Control System**
  - Systems that automatically control spacecraft based on position on the track or elapsed time is not allowed (all control must occur through controller input).